

Appendix A — Compliance and judging map

This appendix is limited to ten pages and does not extend the narrative. The application uses a one-page cover and fifteen narrative pages with the exact official section headings.

Compliance status is explicit: one-inch margins, 11-point body, 9-point tables, accessible Word/PDF output, no government seal, and descriptive public links. Required applicant contact fields remain visibly marked because they were not supplied.

Criterion	Primary application pages	Evidence
Need and solution	1–4	Synthetic case, CCO design, caregiver-evidence gate
Implementation	5–8	Architecture, safety, evaluation, pilot gates
Usability/integration	9–11	Responsive product, a11y, workflow adapters
AI principles	12–13	Agency, transparency, privacy, safety, fairness
Merit/eligibility	14–15	TRL-3 artifacts, prize use, sustainability
Boundary. Appendix evidence does not cure the missing real caregiver record or partner commitment.		

Appendix B — System architecture and trust boundaries

Source artifacts enter an immutable source store. Candidate interpretation is separated from deterministic mission controls. Commitments, permissions, audit events, and workflow adapters remain distinct. An adapter cannot change authority merely because it can reach an external system.

Trust boundaries include household tenancy, purpose-specific disclosure, high-risk professional review, caregiver approval, external acknowledgement, and completion evidence. Outages preserve open work rather than producing a success state.

Boundary	Control	Phase 2 verification
Source → interpretation	Provenance and injection quarantine	Adversarial document tests
Interpretation → activation	Safety level + human approval	Clinical governance sign-off
Household → helper	Purpose, fields, expiry, revocation	Identity and consent testing
HEARTH → external system	Prepare/approve/send/outcome states	Sandbox fault injection

Appendix C — Care commitment object and lifecycle

The 26 synthetic care commitment objects include responsibility, owner, alternates, source, location, excerpt, date, verification class, confidence, due window, dependencies, equipment, skill, privacy, consent, risk, safety level, approval, completion criteria, evidence, backup, escalation, state, duration, and history.

Valid transitions are explicit. Completion and verification require evidence. Superseded objects cannot reactivate. A sent request ordinarily moves to Awaiting external response, not Completed.

Lifecycle example	Allowed next state	Required evidence
Awaiting acceptance	Accepted or reassigned	Helper response
In progress	Awaiting external response	Request sent + recipient
Awaiting external response	Completed	Acknowledgement/outcome
Completed	Verified	Reviewer verification
Superseded	None	Immutable history only

Appendix D — Controlled validation

Smart 40 and the focused 60 are deterministic, consecutive, synthetic checks. They are valuable for reproducibility but intentionally narrow. They do not justify a real-world precision, recall, fairness, clinical, or caregiver-outcome claim.

The suite retains full per-case input, expected behavior, actual output, pass state, model/configuration, timestamp, human-review flag, and correction field in the Data Output Logs.

Suite	Cases	Pass	Key boundary
Smart 40	40	40	Messy input, injection, uncertainty, operational failure
Focused benchmark	60	60	Provenance, conflict, privacy, capacity, lifecycle
Required abstention examples	4	4	No unsupported high-risk action

Appendix E — Locked external-style holdout

The 20 cases and ground truth were hashed before the first execution. The run passed 15/20 and retained five failures with no selected rerun.

Mitigations are stable person identifiers and disambiguation, locale confirmation for numeric dates, terminology-aware escalation, conflict-preserving concurrent correction, and exception-aware recurring schedules. The latter three are release gates when safety or action semantics are affected.

Failure	Observed behavior	Release implication
Duplicate name	First-match assignment	Require stable identity
Ambiguous date	US-locale parse	Require locale confirmation
Clinical shorthand	Generic needs review	Make professional escalation explicit
Conflicting corrections	Last write wins	Persist unresolved conflict
Recurring exception	Daily schedule only	Model exception rule

Appendix F — Timed burden study

The browser contains an empty, resettable study mode for manual and HEARTH-assisted conditions. It records eight fixed tasks, per-step and total time, interactions, help, corrections, confidence, effort, optional de-identified feedback, and a local JSON export.

No participant record is preloaded; no result is transmitted. The proposed design is within-participant and counterbalanced. Correct completion time is primary, with safety and permission errors as guardrails.

Measure	Phase 1 value	Phase 2 analysis
Participants	0	5–8 formative; larger pilot as governed
Time saving	Not measured	Median paired difference + range
Interactions/help/corrections	Instrument ready	Paired and task-level analysis
Confidence/effort	Instrument ready	Distribution and disconfirming cases

Appendix G — Safety, privacy, and security

The safety model separates H0 administrative support, H1 reviewable logistics, H2 sensitive/high-consequence preparation, H3 professional review, and H4 emergency boundaries. Conflicts, uncertainty, and stale sources remain visible.

Privacy controls deny cross-household, revoked, expired, or purposeless access and disclose only allowed fields. Security and compliance remain pre-production work; the appendix does not imply certification.

Threat	Phase 1 control	Production gate
Prompt injection in source	Quarantine as untrusted text	Adversarial monitoring
Over-disclosure	Purpose-specific allowlist	Policy engine + identity
Silent source overwrite	Versions and correction history	Immutable store
False completion	Outcome evidence requirement	Adapter acknowledgements
Audit tampering	Event hash chain	Signed/tamper-evident log

Appendix H — Accessibility evidence

The automated production-build audit covers three key states with Playwright and axe-core, keyboard skip behavior, sampled focus visibility, 320-pixel reflow, mobile navigation, and a token contrast matrix. The final recorded run found zero axe violations in the audited states.

This does not establish conformance or caregiver usability. Screen-reader testing with caregivers, assistive-technology diversity, document reading order in all viewers, and partner accessibility governance remain Phase 2 tasks.

Check	Result	Caveat
Axe: welcome/demo/study	0 violations	Three states only
Keyboard skip	Pass	Automated
320px overflow	Pass	Audited states
Token contrast	AA pairs pass after correction	Sampled tokens
Caregiver screen-reader test	Not performed	Required

Appendix I — Caregiver and partner evidence gates

No consented caregiver research record was supplied. Therefore there is no participant register, empirical theme, authentic quote, caregiver-attributed design change, or measured burden result. The research folder contains transparent empty-state artifacts and a co-implementation plan.

No partner has been contacted or committed through this build. The repository contains a public candidate list, nonbinding letter template, readiness checklist, integration questions, and unresolved-dependency register. Candidates are not represented as relationships.

Evidence class	Current status	Action before claim
Caregiver interview	Missing	Consent + de-identified source + verified analysis
Caregiver quote	Missing	Source check + quote permission
Partner commitment	None	Authorized outreach + written evidence
Integration	Simulation	Sandbox, authority, governance, testing

Appendix J — Cost, sustainability, and roadmap

The six-month pilot estimate is \$145,000 for 100 enrolled households. It includes engineering, implementation, compensated caregiver research, accessibility and translation, clinical/privacy/security advice, cloud/tools, and contingency.

The sustainability strategy protects core caregiver control functions and avoids advertising or sale of personal data. Phase 2 tests buyer value, partner burden, caregiver benefit, equity, and actual unit cost before a pricing commitment.

Milestone	Timing	Decision evidence
Governance + partner scope	Months 1–2	Signed roles, accessible protocol
Safety and platform hardening	Months 2–3	Holdout gates + security review
Formative evaluation	Months 3–5	Usability, burden, guardrails
Pilot decision	Month 6	Caregiver/partner go-no-go